

Wyatt Walsh

SENIOR AI/ML ENGINEER

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Summary: Senior AI/ML engineer shipping production agentic AI systems, LLM document intelligence, RAG/retrieval workflows, AI guardrails, and validated data pipelines and platforms for fintech, risk, compliance, and enterprise tooling.

EXPERIENCE

Senior AI/ML Engineer

Frame Payments | Remote | Oct 2025 - Jan 2026

- Delivered production agentic AI document intelligence, human-in-the-loop review, and multimodal compliance screening for high risk merchant onboarding.
- Deployed production LLM document intelligence with PydanticAI, Amazon Bedrock AgentCore, and Claude; added human-in-the-loop browser feedback for confidential uploads and reduced analyst follow-ups/resubmissions ~90% by blocking miscategorized documents before review.
 - Built multimodal restricted-item compliance screening by combining product images and structured metadata; translated banking-partner policy requirements into automated risk checks for merchant onboarding across tens of merchants and multi-location catalogs.
 - Built a NotebookLM-based technical knowledge hub spanning vendor/web documentation and internal monorepo docs; automated corpus refresh with a Colab pipeline that re-ingests the latest sources to keep retrieval accurate and up to date.
 - Drove internal AI adoption with Notion MCP playbooks, high-signal prompt patterns, and weekly office hours reaching ~15% of the company; standardized PII/confidential-document safeguards and auditability requirements.

Associate III, Senior Software Engineer - Data Integrations

JPMorgan Chase & Co. | New York, NY | May 2023 - Apr 2025

- Built enterprise data pipelines, risk analytics, and data-quality automation for credit trading while scaling secure internal AI developer tooling.
- Pioneered firm-wide rollout of secure generative AI developer tools (LLMs, GitHub Copilot, Codeium) as one of the first 500 engineers, shaping scalable deployment and leading workshops for technical and non-technical teams.
 - Devised ML-driven data-quality anomaly detection for credit trading and risk analytics, reducing false positives by 25% and cutting manual review time so risk teams could focus on critical alerts.
 - Developed resilient Python wrapper for KDB+, democratizing data access for 5,000+ users with CI-style tests and optimized Q functions, reducing manual processing by over 10 hours per week.
 - Engineered a cloud-function-driven asset-fee data sink, consolidating 10+ sources into a centralized database, cutting manual maintenance by 90% and providing on-demand access for traders through a Python SDK, REST API, and interactive dashboard.

Senior Software Engineer

Stealth Web3 Startup | Remote | Mar 2022 - Jul 2022

- Architected a scalable collateralized lending application using React, Next.js, TypeScript, MobX, and ethers.js, translating complex wallet and smart-contract flows into secure, performant user workflows.
- Implemented end-to-end CI/CD with GitHub Actions, Husky, Cypress, and Jest to halve release cycles and raise test coverage to 90%, while mentoring a front-end intern to build key UI components.
- Optimized front-end performance through code splitting, lazy loading, and asset caching, reducing initial load times by 40% and JavaScript bundle size by 30%.

Founder, Chief Technology Officer

SandLabs | Remote | Jun 2021 - Mar 2022

- Engineered and launched a Next.js/TailwindCSS public site with top-tier Lighthouse scores and secured \$75K in grant funding via a competitive ecosystem program.
- Engineered Python pipelines to extract and normalize a 1M+ Reddit posts/comments dataset and publish a reusable data product for external analytics workflows.

Demand Forecasting Extern

Gap Inc. | San Francisco, CA | Jan 2020 - May 2020

- Applied time-series decomposition and bias correction techniques to demand forecasts, boosting accuracy by up to 25% and minimizing surplus inventory costs.
- Owned the data engineering layer by architecting scalable MySQL and Apache Spark pipelines across 5+ years of sales data, improving data processing throughput by 95% and standardizing insights across forecasting teams.

SKILLS

- MACHINE LEARNING AND GENERATIVE AI:** scikit-learn, Hugging Face Transformers, OpenAI API, LangChain, LangGraph, Model Context Protocol (MCP), Retrieval Augmented Generation (RAG), Prompt Engineering, PydanticAI, Amazon Bedrock, Claude, GitHub Copilot, Document AI, AI Guardrails
- DATA ENGINEERING AND ANALYTICS:** Pandas, NumPy, SQLAlchemy, Pydantic, Pandera, Apache Spark, AWS Glue, KDB+, PostgreSQL, DynamoDB, ETL, DuckDB, Polars, Parquet
- PROGRAMMING & SCRIPTING LANGUAGES:** Python, SQL, TypeScript, JavaScript, Go, Rust, Bash
- CLOUD INFRASTRUCTURE AND DEVOPS:** AWS, Docker, Kubernetes, Terraform, GitHub Actions, AWS Lambda, Google Cloud Functions, ECS, EKS

PROJECTS

AI Agent Harness Configs | github.com/wyattowalsh/agents

LLM developer-workflow catalog with 40+ reusable skills, MCP-ready configs, shared instructions, and cross-runtime agent automation.

- Built 40+ reusable instruction and skill packs plus shared harness configs that install across Claude Code, Codex, GitHub Copilot, Cursor, and related agent runtimes.

Tech: Python, TypeScript, MCP, FastMCP

NBA Basketball Database | github.com/wyattowalsh/nbadb

Star-schema analytics warehouse spanning 65K+ NBA games and 13M+ play-by-play events with validated ETL, DuckDB/SQLite/Parquet exports, docs, CLI, and Kaggle distribution.

- Implemented a high-performance, Pandera-validated ETL pipeline that extracts from the NBA API, updates warehouse outputs daily, and exports DuckDB, SQLite, Parquet, and CSV artifacts.
- Built docs, schema references, release workflows, and Kaggle distribution for a 180+ table public warehouse spanning 130+ stats.nba.com endpoint classes.

Tech: Python, Pandas, Polars, SQLAlchemy, DuckDB, Pandera

ProxyWhirl | github.com/wyattowalsh/proxywhirl

Data-collection toolkit with nine rotation strategies, 100+ auto-validated sources, and self-healing proxy health checks.

- Implemented 9 rotation strategies (round-robin, weighted, geo-targeted, performance-based, and more) with configurable fallback chains and per-source health scoring.

Tech: Python, SQLite, PyPI

AI/ML Web Feeds | github.com/wyattowalsh/ai-web-feeds

Typed AI/ML source-discovery platform with HTTP/RSS validation, contributor workflows, OPML exports, and docs-backed feeds for model and agent research.

- Defined a structured YAML feed schema with topic classification (ML, NLP, CV, etc.) and source-type validation (blog, paper, newsletter) to ensure feed quality at scale.

Tech: Python, Pydantic, SQLAlchemy, YAML, Fumadocs, Vercel

MCP-Crawl4AI | github.com/wyattowalsh/mcp-crawl4ai

FastMCP server enabling AI agents to crawl, extract, and return structured web content over documented stdio/HTTP paths.

- Built a FastMCP v3 server with lifespan-managed headless Chromium, session-aware crawling workflows, and automatic Playwright browser setup—only 2 runtime dependencies.

Tech: Python, FastMCP, Crawl4AI, Playwright, Docker

MDXPad | github.com/wyattowalsh/mdxpad

Native macOS MDX editor with synchronized preview, multi-provider AI assistance, and Keychain-backed credential storage.

- Built a dual-format editor on Electron 39 + CodeMirror 6 with synchronized scroll between source and rendered preview, remark/rehype plugin support, and YAML frontmatter validation.

Tech: Electron, React, TypeScript, CodeMirror, Vercel AI SDK

FL Studio MCP Server | github.com/wyattowalsh/fl-mcp

Alpha MCP platform for AI-assisted music production with 190+ typed FL Studio tools, resources-first contracts, transaction-safe mutations, and 650+ tests.

- Defined 190+ FL Studio tools plus core governance tools, with read paths exposed through typed resources and mutations funneled through a transaction pipeline.

Tech: Python, MCP, Next.js, Fumadocs, FL Studio SDK

Personal Website: w4w.dev | github.com/wyattowalsh/personal-website

Long-running personal web app and MDX publishing system pairing a polished frontend with durable content, SEO, and authoring workflows.

- Designed and maintained the public product surface across multiple generations, including responsive landing experiences, theme support, motion, and accessibility-conscious UI polish.

Tech: Next.js, React, TypeScript, Tailwind CSS, MDX

EDUCATION & CERTIFICATIONS

Bachelor of Science

University of California, Berkeley - College of Engineering, Industrial Engineering & Operations Research

Aug 2014 - Dec 2020

High School Diploma

The Hotchkiss School

Aug 2012 - May 2014

CERTIFICATIONS: AWS Certified Cloud Practitioner | IBM Data Science Professional Certificate